WORKED SOLUTION

STEP III 1998, Question 5

Pure • Matrix Exponential

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q5

The key idea

An infinite series of matrices looks frightening, but both matrices here have powers that repeat in a very simple pattern: the powers of $\mathbf{M}$ cycle like the powers of i, and the powers of $\mathbf{N}$ vanish from the square onwards. Once you know every power, the series splits into familiar scalar series (or stops altogether), and you can read off the transformations. The final part is then a direct matrix calculation, and its geometric meaning comes from asking which lines each transformation leaves alone.

Solution

The matrix $\mathbf{M}$

Multiplying out,

$$\mathbf{M}^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -\mathbf{I},$$

as required. Hence, for every integer $k \geq 0$,

$$\mathbf{M}^{2k} = (\mathbf{M}^2)^k = (-1)^k\mathbf{I}, \qquad \mathbf{M}^{2k+1} = (-1)^k\mathbf{M}.$$

Splitting the series into even and odd powers of $\theta\mathbf{M}$,

$$\begin{aligned} \exp(\theta\mathbf{M}) &= \sum_{k=0}^{\infty} \frac{\theta^{2k}}{(2k)!}\mathbf{M}^{2k} + \sum_{k=0}^{\infty} \frac{\theta^{2k+1}}{(2k+1)!}\mathbf{M}^{2k+1} \\ &= \Big(\sum_{k=0}^{\infty} \frac{(-1)^k\theta^{2k}}{(2k)!}\Big)\mathbf{I} + \Big(\sum_{k=0}^{\infty} \frac{(-1)^k\theta^{2k+1}}{(2k+1)!}\Big)\mathbf{M} \\ &= \cos\theta\,\mathbf{I} + \sin\theta\,\mathbf{M} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}. \end{aligned}$$

(Splitting is legitimate because each entry of the matrix series is a series of real numbers whose terms are bounded in size by $|\theta|^r/r!$, so it converges absolutely and may be regrouped.)

Ask yourself Where have I seen $\cos\theta\,\mathbf{I} + \sin\theta\,\mathbf{M}$ before?

Since $\mathbf{M}^2 = -\mathbf{I}$, the matrix $\mathbf{M}$ behaves exactly like i, and the calculation is the matrix version of $e^{i\theta} = \cos\theta + i\sin\theta$. Geometrically, $\exp(\theta\mathbf{M})$ sends $(1,0)$ to $(\cos\theta, \sin\theta)$ and $(0,1)$ to $(-\sin\theta, \cos\theta)$: it is an **anticlockwise rotation of the plane about the origin through angle** θ. (Indeed $\mathbf{M}$ itself is the rotation through $\frac{\pi}{2}$, just as multiplication by i is.)

The matrix $\mathbf{N}$

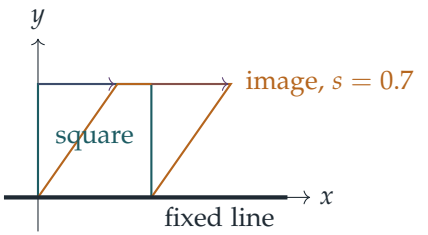
Here

$$\mathbf{N}^2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix},$$

so $\mathbf{N}^r = \mathbf{0}$ for all $r \geq 2$ and the series stops after two terms:

$$\exp(s\mathbf{N}) = \mathbf{I} + s\mathbf{N} = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}.$$

This sends (x, y) to $(x + sy, y)$. Every point moves parallel to the x-axis by a distance proportional to its height y; points on the x-axis do not move at all. So $\exp(s\mathbf{N})$ is a **shear parallel to the x-axis**, with the x-axis as its line of fixed points, sending the unit square to a parallelogram of the same area (the determinant is 1).



When do they commute?

Write $c = \cos\theta$ and $\sigma = \sin\theta$. Then

$$\exp(s\mathbf{N})\exp(\theta\mathbf{M}) = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}\begin{pmatrix} c & -\sigma \\ \sigma & c \end{pmatrix} = \begin{pmatrix} c + s\sigma & -\sigma + sc \\ \sigma & c \end{pmatrix},$$

$$\exp(\theta\mathbf{M})\exp(s\mathbf{N}) = \begin{pmatrix} c & -\sigma \\ \sigma & c \end{pmatrix}\begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} c & sc - \sigma \\ \sigma & s\sigma + c \end{pmatrix}.$$

Subtracting,

$$\exp(s\mathbf{N})\exp(\theta\mathbf{M}) - \exp(\theta\mathbf{M})\exp(s\mathbf{N}) = s\sin\theta\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

This is the zero matrix for every s if and only if $\sin\theta = 0$ (if $\sin\theta \neq 0$, take $s = 1$). So the two matrices commute for all s exactly when

$$\theta = n\pi, \qquad n \in \mathbb{Z}.$$

For these values $\exp(\theta\mathbf{M}) = (-1)^n\mathbf{I}$, which is either the identity or $-\mathbf{I}$.

Ask yourself Why should a rotation that commutes with this shear have to be a half-turn or nothing?

Let R be the rotation and S the shear (with $s \neq 0$), and suppose $SR = RS$. Take any point P on the x-axis, so that $S(P) = P$. Then

$$S(R(P)) = R(S(P)) = R(P),$$

so $R(P)$ is also a fixed point of the shear. But the only points the shear fixes are those on the x-axis. So the rotation must map the x-axis onto itself, and the only rotations about the origin that do this are through 0 (the identity) or π (the half-turn $(x, y) \mapsto (-x, -y)$). Conversely, these two commute with *every* linear map, since they are $\pm\mathbf{I}$: turning the plane through a half-turn and shearing it give the same result in either order. Any other rotation tilts the fixed line of the shear, so shearing before or after rotating gives different results.

Checks

- At $\theta = 0$ the rotation matrix is $\mathbf{I}$, and at $s = 0$ the shear is $\mathbf{I}$, as the series demands.
- Both determinants are 1 ($\cos^2\theta + \sin^2\theta$ and $1 \cdot 1$), as they should be for a rotation and a shear, which preserve area.
- Numerically, summing 80 terms of the series for $\theta = 1$ gives entries matching $\cos 1 \approx 0.5403$ and $\sin 1 \approx 0.8415$. With $\theta = 1$, $s = 1$ the commutator has diagonal entries $\pm\sin 1 \approx \pm 0.8415$, matching $s\sin\theta\,\mathrm{diag}(1, -1)$.

Common slips

- Assuming $\exp(\mathbf{A})\exp(\mathbf{B}) = \exp(\mathbf{A} + \mathbf{B})$ for matrices. That rule needs $\mathbf{AB} = \mathbf{BA}$, which is precisely what fails here unless $\sin\theta = 0$.
- Giving only $\theta = 0$ (or $\theta = 0, \pi$) instead of all integer multiples of π.
- Calling $\exp(s\mathbf{N})$ a stretch or an enlargement: it preserves area and fixes the x-axis pointwise, which is what makes it a shear.

Beyond STEP

The matrix exponential is the bridge between *Lie algebras* and *Lie groups*: $\mathbf{M}$ generates the group of rotations and $\mathbf{N}$ the group of shears, and $\exp(t\mathbf{A})$ solves the linear differential equation $\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x} = \mathbf{A}\mathbf{x}$. The failure of $\exp(\mathbf{A})\exp(\mathbf{B}) = \exp(\mathbf{A} + \mathbf{B})$ for non-commuting matrices is measured by the commutator $\mathbf{AB} - \mathbf{BA}$, an idea you met in the last part. Strang's *Introduction to Linear Algebra* covers matrix exponentials and their use in differential equations.

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